

Curriculum Vitae – Zirui Zhao

CONTACT	School of Computing National University of Singapore 13 Computing Drive Singapore 117417, Republic of Singapore.	Email: zirui@comp.nus.edu.sg GitHub: 1989Ryan Phone: (+65) 9355-3852
EDUCATION	Doctor of Philosophy (Ph.D.) in Computer Science National University of Singapore Advisor: Prof. Wee Sun Lee	Aug 2020 -
	Bachelor of Engineering (B.Eng.) in Automation (EECS) Xi'an Jiaotong University Elite Qian Xuesen Program, excellent graduate.	Aug 2016 - Jun 2020 GPA: 3.92/4.30
VISITING	Carnegie Mellon University , Pittsburgh, U.S.A. <i>Summer Intern</i> , Safe AI Lab The Robotics Institute & Dept. Mechanical Engineering. Advisor: Prof. Ding Zhao	Jul 2019 - Aug 2019
	National University of Singapore , Singapore <i>Visiting Student</i> , 2018 Summer Workshop School of Computing. Advisor: Prof. Soo Yuen Jien	Jul 2018 - Aug 2018
INTERESTS	Robotics; Robot Decision-making under uncertainty; Robot Planning with Large Language Models; Robot Perception; Human-robot interactions.	
SKILLS	Languages: Python, C++, MATLAB, $\LaTeX$.	
	Application Tools: TensorFlow, PyTorch, Keras, Scikit-Learn, Numpy, Jupyter, OpenCV, ROS. Software: Carla, GitLab, GitHub.	
RESEARCH	Planning with Large Language Models <i>Supervisor: Prof. Wee Sun Lee and Prof. David Hsu</i>	2023 - NUS
	We use Large Language Models as both the commonsense model of the world and the heuristic policy within the Monte Carlo Tree Search framework, enabling practical and reasoned decision-making for daily tasks. - LLM's world model provides a commonsense prior belief of states for MCTS to achieve reasoned decision-making efficiently. - The LLM's heuristic policy guides the search to relevant parts of the tree, substantially reducing the search complexity.	
	Grounding Natural Language Instructions for Object Placing <i>Supervisor: Prof. Wee Sun Lee and Prof. David Hsu</i>	2021 - NUS
	We proposed a novel method, ParaGon, for language-conditioned object placing. ParaGon integrates a parsing algorithm into an end-to-end trainable neural network. It is data-efficient and generalizable for learning compositional instructions, and robust to noisy, ambiguous language inputs.	
	Active Risk-sensitive Inverse Reinforcement Learning <i>Supervisor: Ding Zhao</i>	Jul 2019 - Aug 2019 Carnegie Mellon University
	- Active demonstration querying for faster human risk envelope approximation via disturbance planning. - Experimental verification in single-step and multi-step setting with simulated car-following task in Carla.	

Visual Semantic SLAM for Outdoor Environments Nov 2018 - Jun 2019
 Supervisor: Pengju Ren, Cognitive Architecture Group Xi'an Jiaotong University
 - Accomplished visual semantic SLAM based on PSPNet101 and ORB SLAM.
 - The SLAM has implemented with GPS Fusion and topological semantic mapping.
 - GitHub: https://github.com/1989Ryan/Semantic_SLAM

TEACHING **CS 3263 Foundations of Artificial Intelligence** Sem 2 AY22/23 & Sem 1 AY23/24
 Instructor: Prof. Tze-Yun Leong NUS
 - Tutor for course tutorials.
 - Teach topics that cover Logic, Advanced Search, Constraint Satisfaction, Bayesian Networks, Causal Networks, Markov Decision Process, and Reinforcement Learning.

AWARDS

- **Best Paper Runner-up** LTAMP workshop @ Robotics: Science & Systems, 2023
- **NUS Research Scholarship** National University of Singapore, 2020 - 2024
- **Excellent Graduate** Xi'an Jiaotong University, 2020
- **Dean's List** Xi'an Jiaotong University, 2017 & 2018 & 2019
- **2-nd of 1989 Mechanical Alumni Scholarship** Xi'an Jiaotong University, 2018
- **Siyuan Merit Scholarship** Xi'an Jiaotong University, 2017

PUBLICATION

- [4] **Z. Zhao**, W. S. Lee, and D. Hsu. "Large Language Models as Commonsense Knowledge for Large-Scale Task Planning". In: *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS)*. 2023.
- [3] **Z. Zhao**, W. S. Lee, and D. Hsu. "Differentiable Parsing and Visual Grounding of Natural Language Instructions for Object Placement". In: *2023 IEEE International Conference on Robotics and Automation (ICRA)*. 2023, pp. 11546–11553. DOI: 10.1109/ICRA48891.2023.10160640.
- [2] R. Chen, W. Wang, **Z. Zhao**, and D. Zhao. "Active learning for risk-sensitive inverse reinforcement learning". In: *arXiv preprint arXiv:1909.07843* (2019).
- [1] **Z. Zhao**, Y. Mao, Y. Ding, P. Ren, and N. Zheng. "Visual-based semantic SLAM with landmarks for large-scale outdoor environment". In: *2019 2nd China Symposium on Cognitive Computing and Hybrid Intelligence (CCHI)*. IEEE. 2019, pp. 149–154.

REFERENCE

- **Prof. Wee Sun Lee**, Professor @ NUS. leews@comp.nus.edu.sg
- **Prof. David Hsu**, Provost's Chair Professor @ NUS. dyhsu@comp.nus.edu.sg
- **Prof. Ding Zhao**, Associate Professor @ CMU. dingzhao@cmu.edu
- **Prof. Pengju Ren**, Professor @ XJTU. pengjuren@xjtu.edu.cn